ARTI 403 – Image Processing (Project Guidelines)

For this project, students are required to work in teams to develop a research paper and its corresponding implementation. The paper should be prepared in a professional academic format and follow the structure, writing standards, and formatting requirements of a reputable conference in the field of Digital Image Processing.

1. Project Proposal

The first submission, due in **Week 8**, should propose a real-world problem related to the Digital Image Processing domain. The project should be based on relevant research articles studied or reviewed by the group. Submit a brief research proposal containing the following:

A. Project Title
B. Team Members: Include the team leader and all team members.
C. Team Responsibilities: Provide a table showing the assigned responsibilities of each team member.
D. Introduction to the Project Area: Provide a brief introduction to the main image-processing area related to the project.
E. Problem Description: Clearly describe the real-world problem that the project aims to address.
F. Dataset: Provide the link to the dataset that will be used in the project, along with the date on which the dataset was accessed.

2. Manuscript Templates for Writing-up Conference Proceedings

The purpose of a conference template, as specified by the IEEE Manuscript Templates for Conference Proceedings page is to provide you with a consistent format for papers appearing in IEEE conference proceedings. We strongly encourage the use of the conference manuscript templates provided at https://www.ieee.org/conferences/publishing/templates.html

Manuscript templates provide several benefits when drafting papers:

A. Consistency: Using a manuscript template helps to ensure that your paper adheres to the style and formatting guidelines of the conference/journal. This can make it easier for reviewers to focus on the content of your paper rather than being distracted by formatting issues.

B. Ease of use: Manuscript templates typically include pre-formatted sections and styles, making it easier to draft and format your paper. This can save time and reduce the risk of formatting errors.
C. Professional appearance: Using a manuscript template can give your paper a professional appearance and help it stand out among other submissions.
D. Improved readability: Manuscript templates often include suggestions for font type and size, line spacing, and margins, which can improve the readability of your paper.
E. Time-saving: Drafting your paper using a manuscript template can save time as you don't have to worry about formatting and layout issues. This allows you to focus on the content of your paper and get it submitted more quickly.

It is up to the group to decide if they wish to follow the IEEE guidelines when working on this project component. You are hereby free to use any other templates you see fit, and you must mention the source of the templates used as footnotes in your research drafts.

3. Source Code

There are several benefits to publishing the source code used in conference/journal papers:

A. By sharing the source code used in your paper, other researchers can easily reproduce your results and verify the validity of your claims. This can help to increase the transparency and credibility of your work.
B. Sharing your source code can facilitate collaboration with other researchers, allowing them to build upon your work and contribute to its further development.
C. By sharing your source code, you can provide valuable educational resources for other researchers, helping them to learn about new techniques and approaches used in your work.
D. By sharing your source code, you can contribute to the advancement of the field by making your work more widely available and accessible to others.
E. Sharing your source code can also help to increase the impact of your publication by making it easier for others to build upon your work and cite it in their own research.

4. Project Presentation

Students are required to present their project work, explain the key outcomes, and discuss the contributions and responsibilities of each team member. Each team member should be prepared to answer questions related to the project.

5. How to Structure the Paper

The following guidelines outline the required components and recommended structure of your research paper. Students should adapt the structure as appropriate to their project while ensuring that the relevant Digital Image Processing concepts and techniques learned in the course are clearly demonstrated.

I. Introduction

- Introduce the real-world problem addressed by the project.
- Explain the importance and relevance of the problem in the context of Digital Image Processing.
- Provide a brief overview of the proposed solution and the overall approach.

II. Project Goals and Objectives

- Clearly state the main goal of the project.
- Define specific and measurable objectives.
- Describe the image-processing techniques and approaches that will be used to address the problem.
- Identify the expected performance or evaluation criteria.

III. Related Work (Literature Review)

- Review relevant research articles published in reputable academic sources.
- Summarize existing approaches and techniques related to the project.
- Identify limitations, gaps, or challenges in existing work.
- Explain how the proposed project differs from or builds upon existing research.

IV. Data Acquisition and Preprocessing

- Describe the selected dataset, including its source, size, characteristics, and relevant properties.
- Provide the dataset link where applicable.
- Describe the preprocessing steps applied to the images.
- Emphasize the image preprocessing techniques learned in the course where appropriate.

V. Image Processing and Model Development

- Describe the image-processing techniques and algorithms applied to the problem.
- Explain the methodology and overall processing pipeline.
- If a machine learning or deep learning model is used, describe the model architecture and its purpose.
- Describe the training process, if applicable.
- Explain any techniques used to improve the performance of the proposed approach.

VI. Results, Discussion, and Visualization

- Present the experimental results using appropriate tables, figures, and visualizations.
- Evaluate the proposed approach using appropriate performance metrics.
- Discuss and interpret the results in relation to the project objectives.
- Compare the results obtained with relevant existing approaches, where applicable.
- Discuss limitations and possible improvements.

VII. Conclusion and Future Work

- Summarize the main objectives, methodology, and findings of the project.
- Highlight the key outcomes and contributions of the work.
- Discuss potential real-world applications of the proposed approach.
- Provide recommendations for future work and possible improvements.

6. Submitting Your Work

Students are required to submit the research paper and complete implementation of the project by **Week 13**. The submission must include the following:

1. **Research Paper:** Submit the research paper in both **Word and PDF formats**, along with the **plagiarism/similarity report**.
2. **Project Source Code:** Submit the complete source code used to implement the project. The implementation may be submitted as:
   - A Python script (.py), or
   - A Jupyter Notebook (.ipynb) containing all implementation steps in a clear and logical sequence.
   - The code should be well-organized, properly commented, and easy to understand**.**
3. **Presentation Slides:** Submit the PowerPoint presentation that will be used for the in-class project presentation and evaluation.

Note: All submitted files must clearly identify the **project title and team members**.

7. Project Assessment and Grading Rubric

The following table presents the assessment criteria, grading rubric, and required deliverables for the project. Students will be evaluated based on the criteria outlined below.

No.	Assessment Component	Assessment Criteria	Marks	CLO
1	**Project Proposal**	Includes group members, project idea, objectives, related work, and dataset. The proposal is submitted on time and demonstrates effective group planning and coordination.	**2**	**CLO5**
2	**Algorithm Selection and Implementation**	Appropriate image-processing algorithms are selected and implemented correctly. The source code is well-organized, documented, and easy to understand.	**3**	**CLO2**
3	**Research Quality**	The research demonstrates originality and applies appropriate image-processing concepts, methods, and analysis to address the project objectives.	**4**	**CLO3**
4	**Results, Discussion, and Visualization**	Results are clearly presented and appropriately visualized. The findings are logically discussed and communicated in relation to the research objectives.	**2**	**CLO4**
5	**Project Presentation**	Students clearly communicate the project, explain the key findings, and respond effectively to questions related to their work.	**3**	**CLO4**
6	**Conference Style and Formatting**	The group collaboratively prepares the paper according to the selected conference's style and formatting guidelines, including the required manuscript template.	**1**	**CLO5**
	Total		**15**	